

02425 Diffusions and Stochastic Differential Equations

Exercises Week 4: Linear Systems

Viggo Starcke
s250690@dtu.dk

Variance in the Ornstein-Uhlenbeck Process

The objective of this question is to reproduce figure 5.7 in the book (page 108).

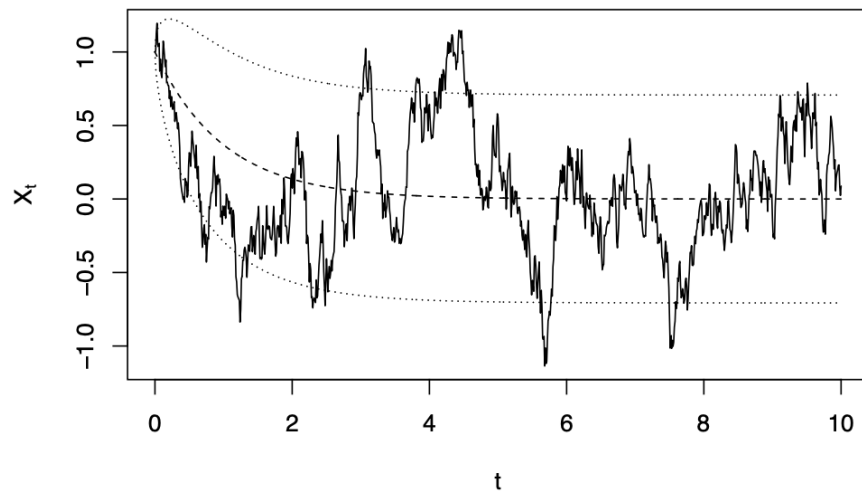


Figure 5.7: A simulation of the Ornstein-Uhlenbeck process (solid) including the expectation (dashed) and plus/minus one standard deviation (dotted). Parameters are $X_0 = 1$, $\lambda = 1$, $\sigma = 1$.

So, consider the scalar linear system

$$\dot{X}_t = -\lambda X_t + \sigma U_t, \quad X_0 = x$$

where $\{U_t : t \geq 0\}$ is Gaussian "white noise", i.e. the formal derivative of standard Brownian motion. The initial condition $X_0 = x$ is deterministic. For numerical work, we take parameters $x = 1$, $\lambda = 1$, $\sigma = 1$.

Question 1 Write up the mean $\mathbb{E}\{X_t\}$ and the variance $\mathbb{V}\{X_t\}$ as a function of time (yes, the expressions can be found in the book). Plot the mean, and the mean plus/minus the standard deviation;

Considering the linear system without white noise, U_t , we have that $-\lambda X_t$ is the expression for the speed of position X_t . Writing it as a differential equation (remember that $\dot{X}_t$ is notation identical to the derivative

wrt time, $\frac{d}{dt}X_t$);

$$\begin{aligned}\dot{X}_t &= -\lambda X_t + \sigma U_t \\ \mathbb{E}\{\dot{X}_t\} &= \mathbb{E}\{-\lambda X_t + \sigma U_t\} \\ \frac{d}{dt}\mathbb{E}\{X_t\} &= -\lambda\mathbb{E}\{X_t\} + \sigma\mathbb{E}\{U_t\} \\ \frac{d}{dt}\mathbb{E}\{X_t\} &= -\lambda\mathbb{E}\{X_t\}\end{aligned}$$

The course literature provides the solution to this derivative as $\mathbb{E}\{X_t\} = e^{-\lambda t}\mathbb{E}\{X_0\}$. For $x = \lambda = \sigma = 1$, we simplify this;

$$\mathbb{E}\{X_t\} = e^{-t}$$

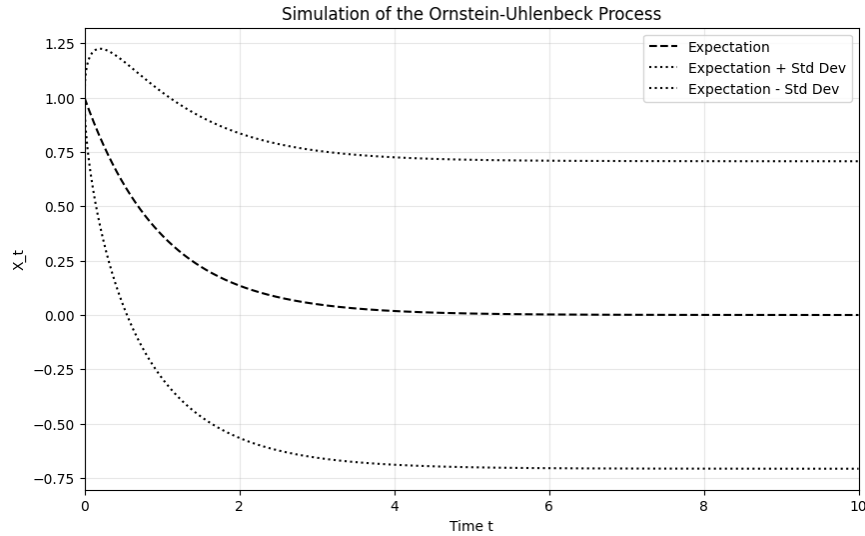
For variance of the Ornstein-Uhlenbeck process the course literature gives the following solution to the variance differential equation;

$$\mathbb{V}\{X_t\} = \frac{\sigma^2}{2\lambda} + \left(\mathbb{V}\{X_0\} - \frac{\sigma^2}{2\lambda}\right)e^{-2\lambda t}$$

Which we simplify for $x = \lambda = \sigma = 1$;

$$\mathbb{V}\{X_t\} = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

Plotting the mean with the upper and lower boundaries plus/minus the standard deviation (the square root of the variance term), we get;



Question 2 Assuming that $\lambda > 0$, write up the steady-state variance, $\lim_{t \rightarrow \infty} \mathbb{V}\{X_t\}$. Compute its numerical value and compare with the value of $\Sigma(t)$ for t large;

For the variance term established in previous question,

$$\mathbb{V}\{X_t\} = \frac{\sigma^2}{2\lambda} + \left(\mathbb{V}\{X_0\} - \frac{\sigma^2}{2\lambda}\right)e^{-2\lambda t}$$

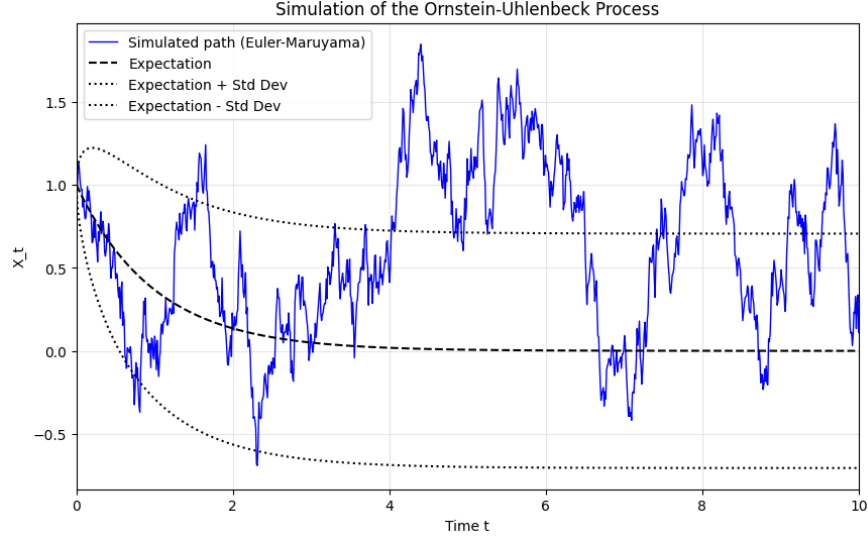
we observe that for $t \rightarrow \infty$ the term goes towards $\mathbb{V}\{X_t\} = \frac{\sigma^2}{2\lambda}$, assuming $\lambda > 0$. This theoretical limit corresponds to the point where the variance from the white noise balances the variance drained by the

system's damping. This equilibrium state is found when solving the Lyapunov equation, which is

$$-2\lambda\Sigma + \sigma^2 = 0$$

for this linear system, where $\lambda = \sigma = 1$. Thus, a numerical solution for the steady-state variance, $\lim_{t \rightarrow \infty} \mathbb{V}\{X_t\}$, is $\Sigma = 0.5$. We compare this with $\Sigma(t)$, which is another way of writing $\mathbb{V}\{X_t\}$, for a large t ; for $t = 10$, we have $\mathbb{V}\{X_t\} = \frac{1}{2} - \frac{1}{2}e^{-2 \cdot 10} \approx \frac{1}{2} - \frac{1}{2}(2 \cdot 10^{-9}) \approx \frac{1}{2}$.

Question 3 Simulate a sample path of $\{X_t\}$ and add it to the graph;



Steady-State Variance Structure for a Mass-Spring-Damper System

Consider the mass-spring-damper system in the notes (5.1), (5.2), page 90, with the force $\{F_t : t \geq 0\}$ being white noise with a given intensity $S_{FF}(\omega) = \sigma^2$.

Question 4 Write the system in the standard form $dX_t = AX_t dt + G dB_t$, i.e. specify A and G . *Hint:* the answer is *almost* given in the textbook. Use that standard Brownian motion is the integral of white noise with intensity 1, and remember that intensity - like variance - is a quadratic property;

From equation (5.4) we have;

$$A = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix} \text{ and } G = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}$$

And from the hint we see that $B_t = \int W_t dt$, which we can take the differential of to get;

$$dB_t = W_t dt \tag{1}$$

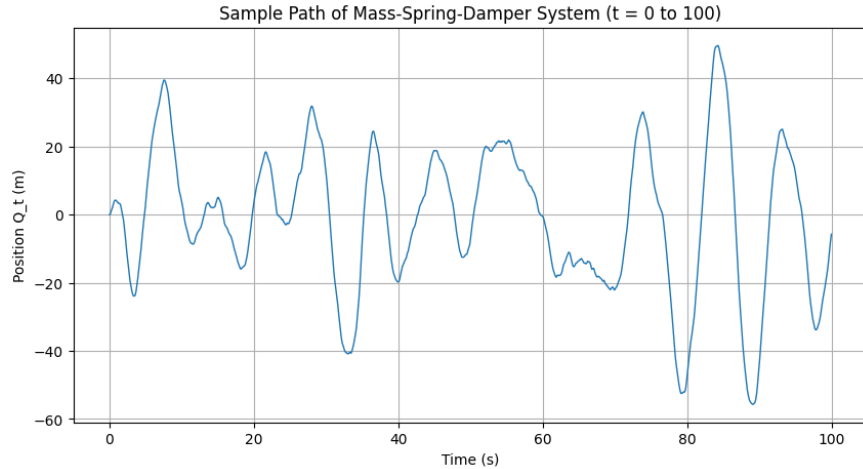
For some standard white noise signal W_t . But we have F_t being our specific white noise force with the intensity $S_{FF}(\omega) = \sigma^2$. Now, since intensity is a quadratic property, the white noise signal F_t is constructed by scaling standard white noise with σ ; $F_t = \sigma W_t$. If we find the differential term on each side we get $F_t dt = \sigma W_t dt$, where we know from (1) that also $F_t dt = dB_t$. Substituting into the standard form;

$$dX_t = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix} X_t dt + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} \sigma dB_t$$

Absorbing the scalar σ into the G gives us back the standard SDE form $dX_t = AX_t dt + G dB_t$, with the resulting matrices;

$$A = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix} \text{ and } G = \begin{bmatrix} 0 \\ \sigma/m \end{bmatrix}$$

Question 5 Using the general form, simulate the system on the time interval $t \in [0, 1000]$ using the Euler-Maruyama method. Take system parameters $m = 1\text{kg}$, $k = 0.5\text{N/m}$, $c = 0.2\text{Ns/m}$, $\sigma^2 = 100\text{N}^2\text{s}$. Let the system start at rest at $t = 0$. Use a time step of $\Delta t = 0.01\text{s}$. Plot the first part of the sample path, $t \in [0, 100]$;



Question 6 Estimate from your simulation the steady-state variance of position Q_t , of velocity V_t , and the covariance between the two. Compare with the solution of the algebraic Lyapunov equation governing the variance. *Note:* In MATLAB, use built-in `lyap.m`. In R, use the function `lyap.R` in `SDEtools`. In Python, use `scipy.linalg.solve_continuous_lyapunov`;

We estimate the steady-state variance in position and velocity aswell as the covariance between the two using `np.cov()` on the second half of our simulation to avoid burn-in variance. For the analytical solution of the Lyapunov equation we exploit the fact that a linear stochastic system $dX_t = AX_t dt + G dB_t$ has the steady-state variance as

$$A\Sigma + \Sigma A^T + GG^T = 0$$

For which Σ is the covariance matrix. Looking at the SciPy documentation for `scipy.linalg.solve_continuous_lyapunov(a, q)` we see it solves the equation formatted as

$$aX + Xa^H = q$$

where a^H is identical to the regular transpose as long as a does not contain any complex numbers. Thus, a solution to the Lyapunov equation is formulated as

$$A\Sigma + \Sigma A^T = -GG^T$$

The results are presented;

Metric	Theoretical	Empirical	Absolute Error
Position Variance	500.00	529.28	29.28
Velocity Variance	250.00	249.46	0.54
Covariance	0.00	-1.20	1.20

The covariance is very low in our Euler-Maruyama simulation, and 0 from the Lyapunov solution. This confirms that at the steady-state, the position and velocity are uncorrelated. We can not predict how fast some mass is moving from knowing its exact position.

Question 7 The kinetic energy is $\frac{1}{2}mV_t^2$ while the potential energy is $\frac{1}{2}kQ_t^2$. In steady-state, what is the expected kinetic energy and the expected potential energy? *Note:* the result is an example of equipartitioning of energy, a general principle in statistical mechanics, both quantum and classical;

We want to find $\mathbb{E}[\frac{1}{2}mV_t^2]$ and $\mathbb{E}[\frac{1}{2}kQ_t^2]$. And since m , k , and $\frac{1}{2}$ are all constants we can pull them out;

$$\begin{aligned}\mathbb{E}[\frac{1}{2}mV_t^2] &= \frac{1}{2}m\mathbb{E}[V_t^2] \\ \mathbb{E}[\frac{1}{2}kQ_t^2] &= \frac{1}{2}k\mathbb{E}[Q_t^2]\end{aligned}$$

Now, for any random variable X we have the variance defined as

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

In our linear system, the mass oscillates symmetrically around zero, so we have $\mathbb{E}[Q_t] = \mathbb{E}[V_t] = 0$. That means;

$$\begin{aligned}\text{Var}(V_t) &= \mathbb{E}[V_t^2] - (0)^2 \\ &= \mathbb{E}[V_t^2]\end{aligned}$$

Thus, we have that the squared velocity is equal to the steady-state velocity variance. And the same logic applies to the position, making $\mathbb{E}[Q_t^2] = \text{Var}(Q_t)$. From the previous question, we have that for the steady-state the variance of the velocity is $\text{Var}(V_t) = 250$, and the variance of the position is $\text{Var}(Q_t) = 500$. Thus, the expected kinetic energy is

$$\frac{1}{2}m\mathbb{E}[V_t^2] = \frac{1}{2}(1)(250) = 125$$

for $m = 1$ (given in question 5). And the expected potential energy is

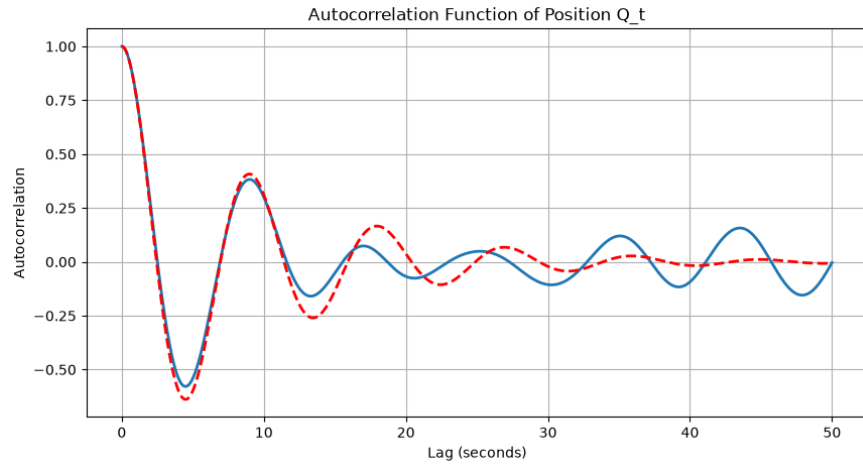
$$\frac{1}{2}k\mathbb{E}[Q_t^2] = \frac{1}{2}(0.5)(500) = 125$$

for $k = 0.5$ (given in question 5). Since both expectations are 125, we confirm this as an example of the equipartitioning of energy. In steady-state, the thermal energy injected by the random noise distributes itself perfectly equally between the kinetic energy (motion) and potential energy (spring tension) of the system.

Question 8 For the simulation, compute and plot the empirical acf of $\{Q_t\}$ up to lag 50s. *Hint:* in MATLAB and R, use `acf`. In Python, use e.g. `statsmodels.tsa.stattools.acf`. Add to the plot the theoretical prediction.;

The theoretical prediction is derived from expressing the deterministic decay over a time delay τ as $e^{A\tau}$. Thus, the theoretical autocovariance matrix at any lag τ is found by multiplying this matrix exponential by our theoretical steady-state covariance matrix Σ ;

$$C(\tau) = e^{A\tau}\Sigma$$



Question 9 Plot, as a function of the frequency ω , the amplitude and phase of the frequency response from the noise to the position. Plot also the theoretical variance spectrum of the position;

